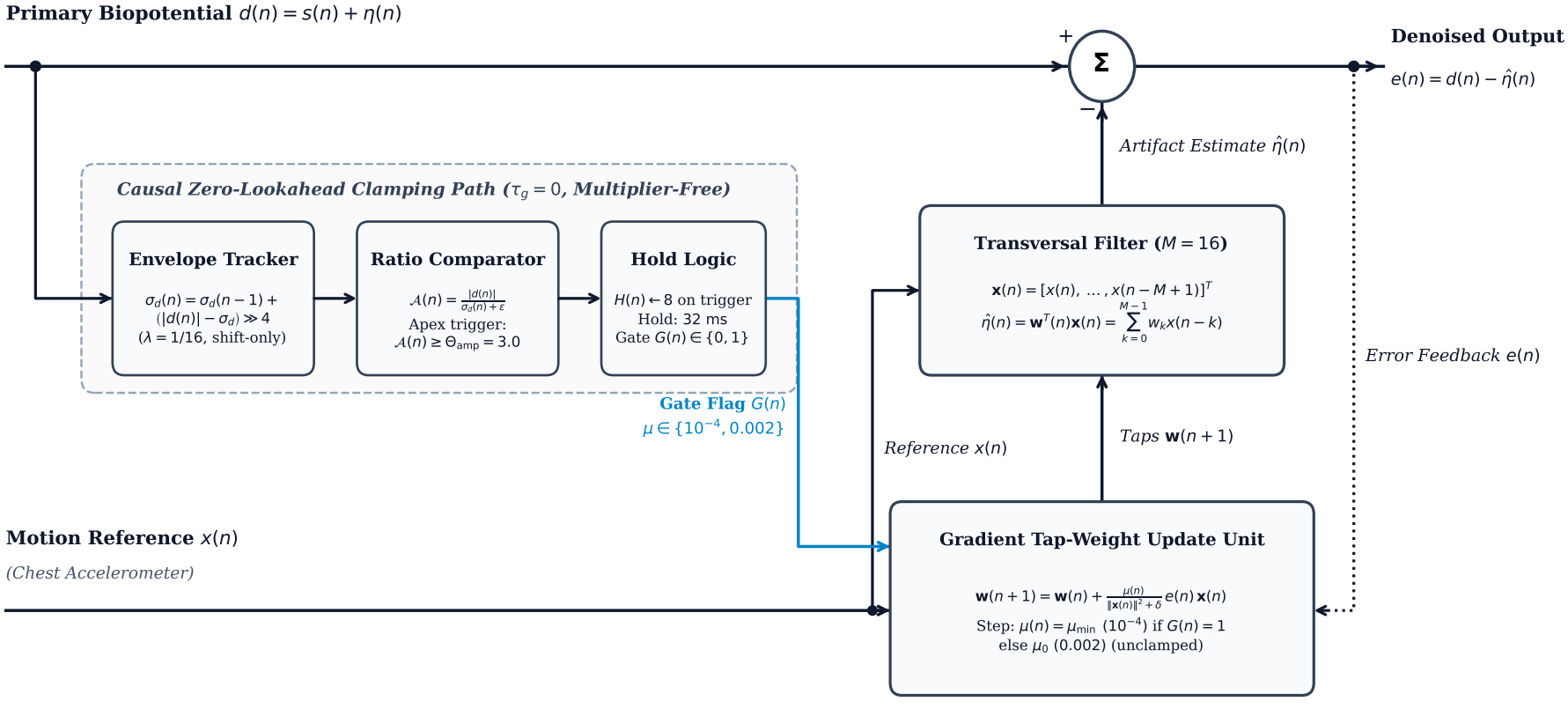

Primary Biopotential $d(n) = s(n) + \eta(n)$
+
Σ
−
Denoised Output
$e(n) = d(n) - \hat{\eta}(n)$
Artifact Estimate $\hat{\eta}(n)$
Causal Zero-Lookahead Clamping Path ($\tau_g = 0$, Multiplier-Free)
Envelope Tracker
$\sigma_d(n) = \sigma_d(n-1) +$
$(|d(n)| - \sigma_d) \gg 4$
($\lambda = 1/16$, shift-only)
Ratio Comparator
$\mathcal{A}(n) = \frac{|d(n)|}{\sigma_d(n) + \varepsilon}$
Apex trigger:
$\mathcal{A}(n) \geq \Theta_{\text{amp}} = 3.0$
Hold Logic
$H(n) \leftarrow 8$ on trigger
Hold: 32 ms
Gate $G(n) \in \{0, 1\}$
Gate Flag $G(n)$
$\mu \in \{10^{-4}, 0.002\}$
Transversal Filter ($M = 16$)
$\mathbf{x}(n) = [x(n), \ldots, x(n-M+1)]^T$
$\hat{\eta}(n) = \mathbf{w}^T(n)\mathbf{x}(n) = \sum_{k=0}^{M-1} w_k x(n-k)$
Error Feedback $e(n)$
Reference $x(n)$
Taps $\mathbf{w}(n+1)$
Motion Reference $x(n)$
(Chest Accelerometer)
Gradient Tap-Weight Update Unit
$\mathbf{w}(n+1) = \mathbf{w}(n) + \frac{\mu(n)}{\|\mathbf{x}(n)\|^2 + \delta} e(n)\mathbf{x}(n)$
Step: $\mu(n) = \mu_{\min}$ (10^{-4}) if $G(n) = 1$
else μ_0 (0.002) (unclamped)